Name: _____ Register No: _____

Experiment 17

Question: Experiment to understand Linear Regression for a given data set.

Aim

To understand the Linear Regression algorithm and predict a continuous value (marks) from study hours.

Code

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

def exp_17():
    from sklearn.linear_model import LinearRegression
    from sklearn.model_selection import train_test_split
    from sklearn.metrics import mean_squared_error, mean_absolute_error, r2_score

    # Data: study hours (X) vs exam marks (y)
    rng = np.random.default_rng(3)
    hours = rng.uniform(1, 10, 100)
    marks = 35 + 5.5 * hours + rng.normal(0, 4, 100)
    df = pd.DataFrame({"Hours": hours.round(2), "Marks": marks.round(2)})
    print("First 5 rows:\n", df.head(), "\n")
    print("Correlation between Hours and Marks:", round(df["Hours"].corr(df["Marks"]), 4), "\n")

    X = df[["Hours"]]
    y = df["Marks"]
    X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
    print("Train size:", len(X_train), "| Test size:", len(X_test))

    model = LinearRegression()
    model.fit(X_train, y_train)
    print("\nSlope (coefficient)  :", round(model.coef_[0], 4))
    print("Intercept            :", round(model.intercept_, 4))
    print(f"Equation             : Marks = {model.intercept_:.2f} + {model.coef_[0]:.2f} x Hours")

    y_pred = model.predict(X_test)
    mse = mean_squared_error(y_test, y_pred)
    print("\nMean Absolute Error  :", round(mean_absolute_error(y_test, y_pred), 4))
    print("Mean Squared Error   :", round(mse, 4))
    print("Root MSE             :", round(np.sqrt(mse), 4))
    print("R-squared (R2) score :", round(r2_score(y_test, y_pred), 4))

    new = pd.DataFrame({"Hours": [4.5, 8]})
    print("\nPrediction for 4.5 hours:", round(model.predict(new)[0], 2))
    print("Prediction for 8 hours  :", round(model.predict(new)[1], 2))

    plt.figure(figsize=(7, 4.5))
    plt.scatter(X_train, y_train, color="steelblue", label="Train data")
    plt.scatter(X_test, y_test, color="orange", label="Test data")
    line_x = np.linspace(1, 10, 50).reshape(-1, 1)
    plt.plot(line_x, model.predict(pd.DataFrame(line_x, columns=["Hours"])), color="red",
             linewidth=2, label="Regression line")
    plt.title("Linear Regression: Study Hours vs Marks")
    plt.xlabel("Hours studied")
    plt.ylabel("Marks")
    plt.legend()
    plt.grid(alpha=0.3)
    plt.show()


exp_17()
```

Output

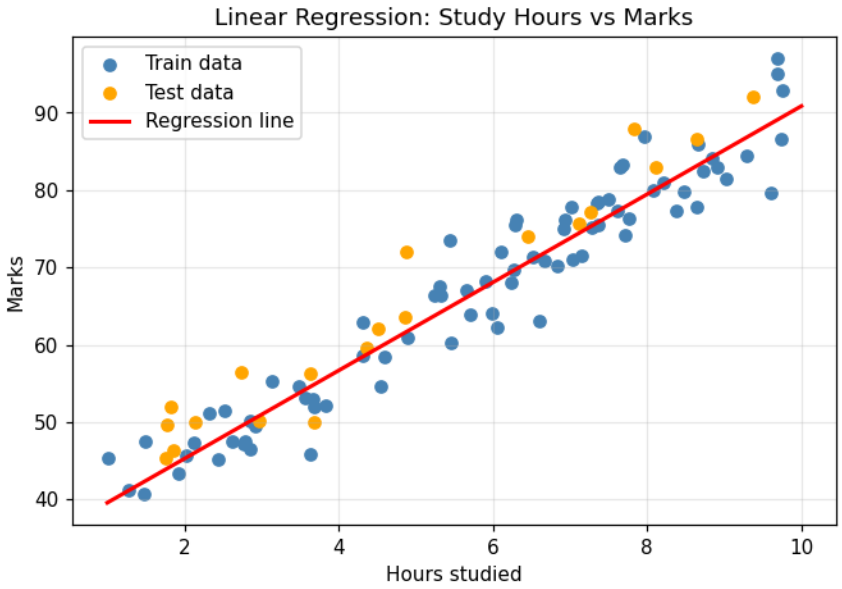
```
First 5 rows:
    Hours  Marks
0   1.77  49.59
1   3.13  55.25
2   8.21  81.03
3   6.24  68.05
4   1.85  46.33
```

```
Correlation between Hours and Marks: 0.9631

Train size: 80 | Test size: 20

Slope (coefficient)  : 5.7005
Intercept            : 33.8646
Equation             : Marks = 33.86 + 5.70 x Hours

Mean Absolute Error  : 3.8703
Mean Squared Error   : 22.7122
Root MSE             : 4.7657
R-squared (R2) score : 0.8975

Prediction for 4.5 hours: 59.52
Prediction for 8 hours  : 79.47
```

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